

Mattex: Supporting transparent and efficient construction material procurement through the cloud

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ABOUT MATTEX



About Mattex

To offer transparent and efficient construction material procurement management services, Mattex leverages automated and secure infrastructure of the Cloud to build a

Founded in 2017, Mattex is a subsidiary of leading Hong Kong construction company, Asia Allied Infrastructure, offering digital construction material purchasing services. eMat, its online construction material procurement and management platform, currently supports more than 300 construction projects of more than 1,000 users by providing services ranging from product sourcing and procurement management, to material transport protection.

Industries: Real Estate & Construction

procurement platform
supports exponential
growth while enhancing
development efficiency

Location: Hong Kong

Storage (<https://cloud.google.com/storage/>),

Google Cloud (<https://cloud.google.com/>)

(<https://cloud.google.com/kubernetes/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Kubernetes (<https://cloud.google.com/kubernetes/>)

Google Cloud results

- Enables app feature release in less than one hour for in-time bug fixes with GKE
- Enhances development efficiency through automatic data backup supported by Cloud SQL
- Shortens feature development time by half with APIs provided by

Support
user growth in
one month

Contact us (</contact>)

Cloud Storage

Digital transformation is expected to bring revolutionary changes to the construction industry. Performing most of their work offline and relying heavily on physical labor, many construction companies are used to procuring and managing their materials with paper-based records, which complicates communication with suppliers and increases the risk of human errors.

Mattex (<https://www.mattex.com.hk/>) aims at helping construction companies operate more efficiently and cost-effectively by leveraging the latest technological solutions to digitally transform the industry. As a subsidiary of leading Hong Kong construction company, Asia Allied Infrastructure, Mattex has been offering digital construction material purchasing services to construction companies, sub-contractors and material suppliers since 2017. Dennis Cheung, Technology Director at Mattex, notes that the construction industry in Hong Kong is only at the beginning of digitalization, with massive room for improvement.

"We believe that technology can help constructors increase their profit margins significantly by optimizing the way they operate. But for a very traditional industry like construction, adopting new changes is not always easy," he adds. "To convince more construction companies to embark on digital transformation, we need to offer reliable solutions that can bring instant positive impact."

In late 2018, Mattex decided to build an online platform that supports transparent and efficient construction material procurement and management. The whole procurement process can be executed online, and all parties are able to easily access the documents they need on any digital device. To ensure high accessibility and scalability, Mattex chose to develop and deploy its platform in a public cloud environment, and landed on Google Cloud (<https://cloud.google.com/>) because it provides intuitive cloud infrastructure tools and a wide range of advanced technologies for further application, like data analytics.

"We wanted our material procurement system to keep evolving along with technological advancements, which is why we needed a cloud platform that is easy to use for developers, and provides extensive capabilities beyond infrastructure," explains Cheung. "Our engineers tested different cloud providers and found that Google Cloud offers the best user experience and has excellent technologies, such as big data and machine learning, that we can leverage later to expand our services."

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—Dennis Cheung,
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Supporting high scalability and quick feature release with GKE

In early 2021, Mattex launched its construction material procurement and management application named eMat, which is run with microservices on Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE). The application began by supporting four construction projects, and has gone through exponential growth to reach more than 300 projects by October 2022. Cheung says that the high scalability of GKE has helped the Mattex team smoothly tackle the fast user growth, with the highest increase of six times (6X) in one month. Now, more than 1000 daily active users of eMat can source products, sign purchase contracts and track orders on the platform any time without technical disruptions, which helps shorten the procurement process time by one third compared to paper-based transactions.

Besides offering reliable infrastructure, GKE also enables Mattex to better manage and improve its services. On GKE, the Mattex team can separate computing resources based on projects, which makes it easier to scale up resources for certain

customers that need it, and view the spendings of each project. On top of that, with the automated deployment feature of GKE, Mattex's DevOps team is able to push code changes to all its microservices on GKE at once, without adjusting them one by one. This means the team can deploy a feature release in less than an hour, allowing Mattex to effortlessly roll out system updates every two weeks and swiftly fix bugs when needed.

"Each construction project has its special needs, so it's common that we have to quickly change codes for certain customers, otherwise their transactions will be blocked," notes Oswald Chan, System Development Manager at Mattex. "By automating deployment workloads, GKE enables us to update our system in minutes and ensure that our customers' projects won't be delayed due to technical bugs."

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*—Oswald Chan, System
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Enhancing development efficiency with Cloud SQL and Cloud Storage

Mattex uses Cloud SQL (<https://cloud.google.com/sql>)
as the database of eMat and stores all the uploaded
files in the system in Cloud Storage
(<https://cloud.google.com/storage>). Apart from
providing fully managed database services and
secure data storage, Cloud SQL and Cloud Storage

have also helped the Mattex team enhance its development efficiency with their extended features. For example, Cloud SQL can automatically back up data, allowing the Mattex team to instantly recover data that are accidentally deleted during the testing stage without the need to recreate them. When the team wanted to develop a feature for eMat to receive large files, it was able to complete the development in a few days by using the API of Cloud Storage that supports the same function. If the team had to develop the feature from scratch, it would have required double the time.

"Cloud SQL and Cloud Storage not only support our platform's daily operations, but also accelerate our development process," says Chan. "Our DevOps team of 15 engineers is able to work more efficiently and continue expanding our services without any overtime."

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Leveraging data analytics and machine learning for higher cost-effectiveness

After successfully building a high-performance platform to facilitate construction material procurement, Mattex plans to leverage more of the latest technologies to help construction companies improve their cost-effectiveness within operations.

The company will use BigQuery

(<https://cloud.google.com/bigquery>) to analyze transaction data for better cost management, and the machine learning tools of Google Cloud to project the trends of construction material prices. At

the same time, it can easily expand to markets outside Hong Kong through the global infrastructure of Google Cloud.

Cheung says, "Transforming a traditional industry like construction is a long journey, and we're still at the very tip of the iceberg. Google Cloud has helped us lay a solid foundation to enable more efficient and transparent construction project management. With the abundant technologies of Google Cloud, we're ready to assist construction companies to further improve their operations."